

```
# coding statistical method.
# student login ID = gg0640
# n+1=0+1 = 1.
```

```
# to perform MEAN in my case .
```

```
import pandas as pd
import numpy as np
```

```
df_variable=pd.read_csv('/content/dataset.csv')
print(df_variable)
```

	State	Year	Nitrogen (%)	Nitrogen (Pounds/Acre)	Phosphorous
0	Alabama	1964	99.0	72.0	100.0
1	Alabama	1965	100.0	81.0	100.0
2	Alabama	1966	100.0	83.0	100.0
3	Alabama	1967	100.0	78.0	100.0
4	Alabama	1968	100.0	71.0	99.0
...	...	...	...	...	...
751	Texas	2013	NaN	NaN	NaN
752	Texas	2014	NaN	NaN	NaN
753	Texas	2015	66.0	65.0	46.0
754	Texas	2016	NaN	NaN	NaN
755	Texas	2017	68.0	77.0	52.0

	Phosphorous (Pounds/Acre)	Potash (%)	Potash (Pounds/Acre)	\
0	62.0	99.0	63.0	
1	63.0	100.0	66.0	
2	69.0	100.0	70.0	
3	71.0	100.0	73.0	
4	71.0	99.0	73.0	
...	...	...	...	
751	NaN	NaN	NaN	
752	NaN	NaN	NaN	
753	34.0	16.0	14.0	
754	NaN	NaN	NaN	
755	37.0	25.0	20.0	

	Area Planted (acres)	Harvested Area (acres)	\
0	NaN	NaN	
1	NaN	NaN	
2	NaN	NaN	
3	NaN	NaN	
4	NaN	NaN	
..	...	...	
751	4800.0	4500.0	
752	5650.0	5200.0	
753	7000.0	5500.0	
754	7750.0	4350.0	
755	7050.0	6000.0	

	Lint Yield (Pounds/Harvested Acre)
0	NaN
1	NaN
2	NaN
3	NaN
4	NaN
..	...
751	610.0
752	748.0
753	809.0
754	756.0
755	568.0

[756 rows x 11 columns]

printing the first five values for each columns
`print(df_variable.head(5))`

	State	Year	Nitrogen (%)	Nitrogen (Pounds/Acre)	Phosphorous (%)	\
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1	Alabama	1965	100.0		81.0	
2	Alabama	1966	100.0		83.0	
3	Alabama	1967	100.0		78.0	
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	Phosphorous (Pounds/Acre)	Potash (%)	Potash (Pounds/Acre)	\
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3	71.0	100.0	73.0	
4	71.0	99.0	73.0	

	Area Planted (acres)	Harvested Area (acres)	\
0	NaN		NaN
1	NaN		NaN
2	NaN		NaN
3	NaN		NaN
4	NaN		NaN

	Lint Yield (Pounds/Harvested Acre)
0	NaN
1	NaN
2	NaN
3	NaN
4	NaN

```
df_variable_one=df_variable['Nitrogen (%)'] # assigning the columns
nitrogen to a new variable which only has a Nitrogen (%) column
df_variable_one.mean() # printing the mean of it .
```

```
89.52136752136752
```

```
df_variable.mean() # printing the mean of each column which contains
numeric values .
```

```
<ipython-input-12-dc3c3f0a3140>:1: FutureWarning: The default value of
numeric_only in DataFrame.mean is deprecated. In a future version, it
will default to False. In addition, specifying 'numeric_only=None' is
deprecated. Select only valid columns or specify the value of
numeric_only to silence this warning.
```

```
df_variable.mean() # printing the mean of each column which contains
numeric values .
```

Year	1990.500000
Nitrogen (%)	89.521368
Nitrogen (Pounds/Acre)	87.205128
Phosphorous (%)	65.450855
Phosphorous (Pounds/Acre)	53.367521
Potash (%)	58.660173
Potash (Pounds/Acre)	58.192140
Area Planted (acres)	874.279365
Harvested Area (acres)	778.571429
Lint Yield (Pounds/Harvested Acre)	772.487302

dtype: float64